

Description

At 9:00 AM the next day, June 18, 2025, the patient is visited by the surgeon and an attending physician who perform a physical to check for signs and symptoms of any post-operation complications. The patient reports that he is experiencing chest pain. The physicians both observe that the patient has signs of cardiac arrhythmia (I49.9). However, the patient suddenly suffers a pulmonary embolism and dies at 12pm. The pulmonary embolism (ICD-10 I26 Pulmonary embolism) is documented as a final diagnosis.

NIST Hospital reports syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) at least once per day. At 12:30 PM on June 18, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT 08 Update message about this encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity.

Comments

The dates and times in this test case illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the test case with an HIT Module, only date and time format will be validated within tester submitted test data.

Pre-condition

A04-Registration message is sent before the A08-Update message.

Post-Condition

No Post-Condition

Test Objectives

This test step examines an HIT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

- Setting is inpatient.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient name is de-identified in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a de-identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Addition of Diagnoses
- Patient Expired

Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

DG1(1)-3

Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
I209	Angina pectoris, unspecified	I10C	I209^Angina pectoris, unspecified^I10C	Yes

194828000	Angina (disorder)	SCT	194828000^Angina (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
I20.9	Angina pectoris, unspecified	I10C	I20.9^Angina pectoris, unspecified ^I10C	Yes
I209	Angina pectoris, unspecified	I10	I209^Angina pectoris, unspecified^I10	No
I20.9	Angina pectoris, unspecified	I10	I20.9^Angina pectoris, unspecified^I10	No
69051000119104	Angina pectoris not detected on treadmill stress test (finding)	SCT	69051000119104^Angina pectoris not detected on treadmill stress test (finding)^SCT	Yes
DG1(2)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
I25110	Atherosclerotic heart disease of native coronary artery with unstable angina pectoris	I10C	I25110^Atherosclerotic heart disease of native coronary artery with unstable angina pectoris^I10C	Yes
53741008	Coronary arteriosclerosis (disorder)	SCT	53741008^Coronary arteriosclerosis (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
I25.110	Atherosclerotic heart disease of native coronary artery with unstable angina pectoris	I10C	I25.110^Atherosclerotic heart disease of native coronary artery with unstable angina pectoris^I10C	Yes
I25110	Atherosclerotic heart disease of native coronary artery with unstable angina pectoris	I10	I25110^Atherosclerotic heart disease of native coronary artery with unstable angina pectoris^I10	No
I25.110	Atherosclerotic heart disease of native coronary artery with unstable angina pectoris	I10	I25.110^Atherosclerotic heart disease of native coronary artery with unstable angina pectoris^I10	No
39839004	Diaphragmatic hernia (disorder)	SCT	39839004^Diaphragmatic hernia (disorder)^SCT	Yes
DG1(3)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
I499	Cardiac arrhythmia, unspecified	I10C	I499^Cardiac arrhythmia, unspecified^I10C	Yes
698247007	Cardiac arrhythmia (disorder)	SCT	698247007^Cardiac arrhythmia (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
I49.9	Cardiac arrhythmia, unspecified	I10C	I49.9^Cardiac arrhythmia, unspecified ^I10C	Yes

I499	Cardiac arrhythmia, unspecified	I10	I499^Cardiac arrhythmia, unspecified^I10	No
I49.9	Cardiac arrhythmia, unspecified	I10	I49.9^Cardiac arrhythmia, unspecified^I10	No
44103008	Ventricular arrhythmia (disorder)	SCT	44103008^Ventricular arrhythmia (disorder)^SCT	Yes
DG1(4)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
I2699	Other pulmonary embolism without acute cor pulmonale	I10C	I26.99^Other pulmonary embolism without acute cor pulmonale^I10C	Yes
233936003	Acute massive pulmonary embolism (disorder)	SCT	233936003^Acute massive pulmonary embolism (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
I26.99	Other pulmonary embolism without acute cor pulmonale	I10C	I26.99^Other pulmonary embolism without acute cor pulmonale^I10C	Yes
I2699	Other pulmonary embolism without acute cor pulmonale	I10	I2699^Other pulmonary embolism without acute cor pulmonale^I10	No
I26.99	Other pulmonary embolism without acute cor pulmonale	I10	I26.99^Other pulmonary embolism without acute cor pulmonale^I10	No
706870000	Acute pulmonary embolism (disorder)	SCT	706870000^ Acute pulmonary embolism (disorder)^SCT	Yes

The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues and error.

Notes

ICD-10 CM codes are provided in the Test Data and required for coding the diagnoses in this Test Case. If the HIT developer uses different but equivalent ICD-10 CM codes than the ones provided, the Tester may ignore errors generated by the Test Tool related to incorrect code when the ICD-10 CM code data used in the message are determined via visual inspection to be valid data.

ICD-10 CM codes are acceptable with or without decimals.

Messages in this test case will be used in a Manual Test Step under functional requirement testing.